

Six Mathematical Models of Animal Navigation, Movement, and Homing: A Unified Survey

TL;DR

- The field's six most-adopted modeling families are: (1) Correlated/Biased Correlated Random Walks (CRW/BCRW), (2) Lévy walks, (3) State-space/Hidden Markov models including the continuous-time CRW (CTCRW) and OU/OUF home-range models, (4) Step-Selection Functions / integrated step-selection analysis (iSSA), (5) Vector navigation / path integration (Müller-Wehner, Mittelstaedt, Mouritsen clock-and-compass), and (6) Cognitive/learning models (reinforcement learning, the successor representation, Bayesian cue integration, and Gaussian-process route models).
- They form a single connected family tree: a discrete random walk is the Euler discretization of a velocity-jump SDE whose diffusion limit is the Fokker-Planck/advection-diffusion PDE; HMMs/SSMs generalize random walks by adding a latent behavioral state and measurement-error layer; SSFs embed the CRW movement kernel inside a Gibbs/Boltzmann utilization distribution driven by habitat; vector-navigation supplies the drift term (bias) that turns a CRW into a BCRW; and RL/Bayesian models supply the value gradients and posterior means that *are* that drift field.
- For a computational cognitive scientist: treat the whole literature as inference over a state-space model whose transition density is a (possibly state-switching, habitat-modulated) biased correlated random walk, whose drift is a policy/value gradient or a Bayesian posterior mean over a goal, and whose diffusion limit connects individual-step mechanism to population-level redistribution.

Key Findings

- **CRW/BCRW is the foundational primitive.** Movement is decomposed into step lengths (gamma/Weibull/exponential) and turning angles (von Mises/wrapped Cauchy). The Kareiva & Shigesada (1983) mean-squared-displacement formula is the canonical diagnostic; the central limit theorem gives a Gaussian/diffusion limit for finite-variance steps. Homing ("map-and-compass") is captured by adding a directional bias toward the goal, giving the BCRW.
- **Lévy walks** replace finite-variance steps with a heavy-tailed power law $P(l) \propto l^{-(\mu+1)}$, $1 < \mu \leq 3$, producing superdiffusion. The Lévy flight foraging hypothesis ($\mu \approx 2$ optimal) is influential but contested; the Edwards et al. critique forced the field to adopt maximum-likelihood + AIC model selection, and many earlier "Lévy" claims were overturned or re-explained as composite CRWs.
- **State-space/HMM models** are now the workhorse for telemetry. moveHMM/momentuHMM fit behavioral-state-switching CRWs via the forward algorithm;

the CTCRW (Johnson et al. 2008, crawl) models velocity as an Ornstein-Uhlenbeck SDE and fits via the Kalman filter, handling irregular sampling and measurement error; the ctm package's OU/OUF processes plus autocorrelated kernel density estimation (AKDE) give statistically rigorous home ranges.

- **Step-selection functions** unify movement and habitat selection as a conditional-logistic / inhomogeneous Poisson point-process likelihood, with a selection-free movement kernel (the CRW) multiplied by an exponential habitat-selection kernel. iSSA (Avgar et al. 2016) is mathematically equivalent to a BCRW and connects to the Boltzmann/Gibbs utilization distribution.
- **Vector navigation / path integration** is the dominant mechanistic account of insect homing (desert ants, bees) and naïve bird migration. The home vector is accumulated by integrating compass heading and distance; Müller & Wehner (1988) showed ants use a computationally cheap *approximation* with systematic errors, while Mouritsen & Mouritsen (2000) showed a wrapped-Cauchy clock-and-compass model predicts a parabolic spread of ringing recoveries.
- **Cognitive/learning models** answer "where to go": RL value functions and policies, the successor representation (a predictive map linked to place/grid cells), Bayesian optimal cue integration (inverse-variance weighting of compasses/path integration), and Gaussian-process models of habitual routes (Mann et al. 2011 homing pigeons). These supply the goal-directed drift that mechanistic movement models take as given.

Details

Model 1 — Correlated and Biased Correlated Random Walks (CRW / BCRW)

(a) Core idea. The CRW models a path as a sequence of steps with *directional persistence*: each heading is correlated with the previous one. It is the null model of "how an animal moves" once you accept that animals tend to keep going forward. The BCRW adds a global bias toward a goal (home, a reef, a resource), making it the canonical mechanistic model of Kramer's "map-and-compass" homing: the *map* sets the preferred direction, the *compass* executes it, and persistence plus noise produce realistic tortuosity.

(b) Key mathematics. A 2-D discrete walk: position $\mathbf{x}_t = \mathbf{x}_{t-1} + l_t(\cos \phi_t, \sin \phi_t)$. Step lengths l_t are drawn from a positive distribution — commonly gamma, Weibull, or exponential. Turning angles (CRW) or absolute headings (BRW) are drawn from a circular distribution — the **von Mises** density $f(\theta; \mu, \kappa) = \exp(\kappa \cos(\theta - \mu)) / (2\pi I_0(\kappa))$, or the **wrapped Cauchy** $f(\theta; \mu, \rho) = (1/2\pi)(1 - \rho^2)/(1 + \rho^2 - 2\rho \cos(\theta - \mu))$. The concentration parameter (κ or ρ) tunes persistence.

The central result is the Kareiva & Shigesada (1983) expected net squared displacement after n steps. Writing the moments of step length and the mean cosine $c = E[\cos \phi]$ and sine $s = E[\sin \phi]$ of the turn-angle distribution, the MSD takes the closed form derived in Codling, Plank & Benhamou (2008, their Eq. 3.3), which contains a term in r^{n+1} (the directional correlation at lag $n+1$, $r < 1$). Two limits matter:

- No persistence ($c=s=0$): $MSD \propto n$ (normal diffusion, the simple random walk $\rightarrow$ Brownian motion).
- Strong persistence ($r \rightarrow 1$): $MSD \propto n^2$ (ballistic motion). Real animals lie between these. For finite-variance steps the CLT guarantees that the position converges to a bivariate Gaussian, so the CRW's continuum limit is a diffusion (Fokker-Planck) process; adding bias gives the drift-diffusion (advection-diffusion) equation.

In the BCRW the expected heading at each step is a weighted compromise between persistence (previous heading ψ_{t-1}) and bias (the direction toward the goal). Bailey et al. (2018, Ecology) parameterize this with a weight β : when $\beta=0$ the expected direction equals the previous heading (pure CRW), and as $\beta \rightarrow \infty$ it equals the goal direction (pure BRW). This single parameter interpolates the entire map-and-compass continuum.

(c) Validation / species. Kareiva & Shigesada fit the CRW to *Pieris rapae* (cabbage white butterfly) flight and *Battus philenor* (pipevine swallowtail) crawling. The CRW/BCRW framework has since been applied across insects, marine larvae aiming for reefs (Codling et al. 2004), and as the building block of essentially all multistate movement models. Bovet & Benhamou (1988) established the spatial-analysis methodology.

(d) Software / canonical references. Kareiva & Shigesada (1983, Oecologia 56:234–238); Bovet & Benhamou (1988, J Theor Biol 131:419–433); Turchin (1998, *Quantitative Analysis of Movement*); Codling, Plank & Benhamou (2008, J R Soc Interface 5:813–834) — the canonical review. R packages: trajr (TrajExpectedSquareDisplacement implements the K&S formula), adehabitatLT, and the simulation tools in moveHMM.

(e) Limitations. The CRW is scale-dependent: step-length and turn-angle distributions change with the sampling interval, so parameters are only interpretable relative to that interval. It is non-trivial to distinguish a CRW from a BCRW when individuals have different (unknown) target directions, because local persistence mimics global bias (Benhamou 2006). The discrete-time formulation also assumes regular sampling and ignores measurement error — exactly the gaps that SSMs fill.

Model 2 — Lévy Walks and the Lévy Flight Foraging Hypothesis

(a) Core idea. A Lévy walk is a random walk whose step-length distribution is heavy-tailed (infinite variance), producing clusters of short steps punctuated by rare very long relocations, with scale-invariant (fractal) structure. The Lévy flight foraging hypothesis (LFFH) holds that natural selection favors this strategy when targets are sparse and non-destructive, because it optimizes search efficiency.

(b) Key mathematics. Step lengths obey a power law $P(l) = a \cdot l^{-(1+\mu)}$ with exponent $1 < \mu \leq 3$ (a is a normalizing constant). For $1 < \mu < 3$ the variance diverges, the CLT does not apply, and the walk is **superdiffusive**: $MSD \propto t^\gamma$ with $\gamma > 1$ (versus $\gamma=1$ for Brownian diffusion). Viswanathan et al. (1999, Nature 401:911–914, "Optimizing the success of random searches") proved the central

optimality claim: "when the target sites are sparse and can be visited any number of times, an inverse square power-law distribution of flight lengths, corresponding to Lévy flight motion, is an optimal strategy" — i.e. $\mu \approx 2$, the "Cauchy walk." The competing models in any rigorous test are the truncated power law, exponential, and gamma; selection is by maximum likelihood plus AIC weights, with goodness-of-fit (Kolmogorov–Smirnov) checks.

(c) Validation / controversy. The hypothesis launched with wandering albatrosses (Viswanathan et al. 1996, *Nature*) and fruit flies. Edwards et al. (2007, *Nature*) re-analyzed the albatross data and showed the original Lévy signal was a statistical artifact of binning and inappropriate fitting; the corrected maximum-likelihood approach (Edwards 2008, *J Anim Ecol* 77:1212–1222; Edwards 2011) overturned several headline claims. Some support survives under rigorous methods — e.g., evidence of Lévy-walk movement in Hadza hunter-gatherers (Raichlen et al. 2014, *PNAS*) — but Pyke (2015, *Methods Ecol Evol* 6:1–16) argued for abandoning the hypothesis outright: "the evidence that organisms actually 'do the Lévy walk' is weak to non-existent, despite claims to the contrary... it is therefore time to divorce the Lévy walk model from optimal foraging theory." A key deflationary result: composite CRWs with spatially dependent turning can generate apparent power laws (Benhamou 2007), so a Lévy fit is not evidence of a Lévy mechanism.

(d) Software / references. Viswanathan et al. (1996, 1999); Edwards et al. (2007, *Nature*); Edwards (2008, *J Anim Ecol*); Pyke (2015, *Methods Ecol Evol* 6); the Reynolds review series. Likelihood-fitting code is distributed with these papers and in the *powerlaw* R package ecosystem.

(e) Limitations. Statistical identifiability is the central problem: distinguishing a true Lévy process from a truncated power law, a composite/CRW mixture, or an environmentally-induced pattern requires large clean datasets and careful MLE. Higher-dimensional optimality of $\mu=2$ is disputed (Levernier et al.), and "Lévy-like" pattern at the population level can emerge from among-individual heterogeneity in simple CRWs.

Model 3 — State-Space and Hidden Markov Models (incl. CTCRW, OU/OUF, Brownian bridge)

(a) Core idea. SSMS separate the *state/movement process* (what the animal is doing) from the *observation process* (noisy, irregular telemetry). HMMs are the discrete-state special case: the animal switches between a small number of behavioral states (e.g., "encamped" vs "exploratory"), each generating its own CRW. This is the unifying generalization — random walks become the within-state emission model, and the framework natively handles measurement error and irregular sampling.

(b) Key mathematics. *HMM*: a latent state $S_t \in \{1, \dots, N\}$ evolves as a Markov chain with transition matrix Γ (whose entries can depend on covariates via a multinomial logit link). Conditional on $S_t = j$, the observed step length and turn angle are drawn from state-specific distributions (gamma + von Mises). The likelihood is computed by the forward algorithm: $L = \delta \Gamma P(x_1) \Gamma P(x_2) \dots \Gamma P(x_T) 1'$, where $P(x_t)$ is the diagonal matrix of state-dependent densities; the Viterbi algorithm decodes the most likely state sequence. This is the same machinery as in speech

recognition, applied to step/angle time series (Langrock et al. 2012, Ecology; Morales et al. 2004 introduced the mixture-of-random-walks idea).

CTCRW (Johnson et al. 2008): the velocity v_t is an Ornstein–Uhlenbeck process, $dv_t = \beta(\gamma - v_t)dt + \sigma dW_t$, and position is its integral, $dz_t = v_t dt$ (the "integrated OU" process). Here β is the velocity mean-reversion (autocorrelation) rate, γ the mean velocity (often 0, no drift), σ the diffusion. Because the transition densities of v_t and z_t are available analytically (via Itô calculus), the model is cast as a linear Gaussian state-space model and fit by the Kalman filter, which integrates over both measurement error and unobserved velocities simultaneously. This handles nonuniformly-sampled data without interpolation.

OU/OUF home range (Fleming, Calabrese, ctmm): an OU *position* process has a stationary (range-resident) distribution — its semivariance asymptotes — giving a home range; the OUF ("OU-Foraging") process adds correlated velocities so that both position and velocity are autocorrelated. AKDE then conditions the kernel bandwidth on the fitted autocorrelation model. Noonan et al. (2019, Ecological Monographs 89(2)) — the largest empirical tracking comparison assembled to date, in which 368 of 369 datasets featured strong autocorrelation — found that conventional IID estimators (MCP, KDE, k-LoCoH) underestimate home-range area by a factor of roughly 2 to 13 on average depending on method and bandwidth optimizer, whereas AKDE was "consistently accurate across the full spectrum of sample sizes and autocorrelation strengths."

Brownian bridge (Horne et al. 2007): a conditional random walk between successive fixes, with a Brownian-motion variance parameter σ^2_m estimated by leave-one-out; the dynamic BBMM (Kranstauber et al. 2012) lets σ^2_m vary along the track. BBMMs estimate the *occurrence* distribution (where the animal was during sampling), conceptually distinct from the *range* distribution AKDE targets.

(c) Validation / species. moveHMM/momentuHMM have been applied to a huge range — African wild dogs, seals, fish via acoustic telemetry, and many others. The CTCRW was introduced on harbor seal and northern fur seal telemetry; OUF/AKDE is validated on African buffalo (the canonical ctmm "Cilla" dataset), and BBMMs on caribou, mule deer migration stopovers, bald eagles, and black-bear highway crossings.

(d) Software / references. moveHMM (Michelot, Langrock & Patterson 2016, Methods Ecol Evol 7:1308–1315); momentuHMM (McClintock & Michelot 2018, MEE 9:1518–1530); crawl (Johnson et al. 2008, Ecology 89:1208–1215); ctmm (Calabrese, Fleming & Gurarie 2016, MEE 7:1124–1132; Fleming et al. 2015 Ecology for AKDE; Noonan et al. 2019 Ecol Monogr for the comparative validation); state-switching CTCRW (Michelot & Blackwell 2019, MEE 10:637–649); Patterson et al. and McClintock et al. (2020, Ecology Letters) reviews; Zucchini, MacDonald & Langrock (2017) textbook.

(e) Limitations. HMM outputs are resolution-dependent (the same caveat as the CRW). State number is chosen by the analyst and can be hard to interpret biologically. Standard HMMs assume the discrete-time grid is regular (CTCRW and state-switching CTCRW relax this).

Kalman-filter inference assumes linear-Gaussian structure; strong nonlinearity needs particle filters or MCMC.

Model 4 — Step-Selection Functions and Integrated Step-Selection Analysis (iSSA)

(a) Core idea. SSFs answer "how do habitat and movement *jointly* shape where the animal goes next?" Each observed "used" step is compared against a set of "available" steps radiating from the same start point; conditional logistic regression then estimates how habitat covariates and movement attributes predict the choice. iSSA (Avgar et al. 2016) makes this a full mechanistic movement model by simultaneously estimating the movement kernel and the selection coefficients.

(b) Key mathematics. The next-location density factorizes as a selection-free movement kernel φ times a habitat-selection kernel w : $u(s, t+1) \propto w(x(s) \cdot \beta) \cdot \varphi(s, s'; \gamma) \cdot u(s', t)$, where $w(x \cdot \beta) = \exp(\beta \cdot x(s))$ is the exponential habitat-selection function and φ is the CRW redistribution kernel (gamma step length $\times$ von Mises turn angle). Taking both in exponential form, the maximum-likelihood estimates of β and the movement parameters are obtained by conditional logistic regression (a Cox partial-likelihood / discrete-choice routine), matching each used step to its available set. This is formally equivalent to fitting an inhomogeneous Poisson point process / weighted distribution. Crucially, including movement covariates — log step length, step length, $\cos(\text{turn angle})$ — as predictors lets the regression coefficients *recover the parameters of the underlying step-length and turn-angle distributions*, and interactions with habitat let the movement kernel itself depend on the environment.

The PLOS One result (Avgar's group) makes the connection to Model 1 explicit: the SSF is mathematically equivalent to a BCRW, with the same β -weighted compromise between persistence and bias. And the steady-state utilization distribution implied by an SSF is a Boltzmann/Gibbs distribution $\propto \exp(\beta \cdot x(s))$ over habitat value — connecting to the Langevin/potential-function view where $-\beta \cdot x(s)$ acts as a potential and its gradient is a drift field.

(c) Validation / species. Fortin et al. (2005, Ecology) — elk responding to wolves in Yellowstone — is the foundational empirical SSF. iSSA has since become ubiquitous in large-mammal ecology (the amt "Lupe" example dataset; wolves, elk, grizzly bears, roads/anthropogenic responses). Thurfjell, Ciuti & Boyce (2014, Movement Ecology) review applications.

(d) Software / references. Fortin et al. (2005); Forester et al. (2009, Ecology) — movement-kernel correction; Avgar, Potts, Lewis & Boyce (2016, MEE 7:619–630); Thurfjell et al. (2014); Fieberg et al. (2021, J Anim Ecol) — the "how-to guide"; Signer, Fieberg & Avgar (2019) — the amt R package; Michelot et al. (2024, MEE) — SSF via numerical integration. The $\text{CLR} \leftrightarrow \text{Poisson}$ equivalence (Muff et al.) enables random effects.

(e) Limitations. Inference is conditional on movement and assumes movement is independent of selection within a step, which can bias estimates when habitat strongly shapes movement. The $\text{CLR} \leftrightarrow \text{SSF}$ equivalence holds only for exponential-family kernels (hence the near-universal default of gamma/exponential step lengths + von Mises angles). SSF *coefficients* measure local

selection, not the steady-state space-use distribution directly — that requires simulation. Availability-set definition remains a known sensitivity.

Model 5 — Vector Navigation and Path Integration (Map-and-Compass, Dead Reckoning)

(a) Core idea. This is the dominant *mechanistic, cognitively-explicit* account of homing in insects (desert ants *Cataglyphis*, bees) and of naïve first-migration in birds. The animal continuously integrates its compass heading and distance travelled into a running **home vector** that always points back to the start. Compasses include the time-compensated sun/sky-polarization compass, the magnetic (inclination) compass, and the star compass; odometry comes from optic flow and stride integration ("step counting"). Kramer's map-and-compass concept frames goal-directed navigation as "consult the map to choose a direction, then hold it with a compass." PubMed Central

(b) Key mathematics. Two coordinate schemes dominate (Vickerstaff & Cheung 2012). The **geocentric polar (GP)** home vector (distance r , angle v relative to an external compass) updates, in geometrically-exact continuous form (Merkle, Rost & Alt 2006, Eqs. 1-2; φ = current heading, v = speed): $dr/dt = v \cdot \cos(\varphi - v)$, $dv/dt = (v/r) \cdot \sin(\varphi - v)$. The discrete per-segment form (step δ) is $r_{n+1} \approx r_n + \delta \cdot \cos(\varphi_n - v_n)$; $v_{n+1} \approx v_n + (\delta/r_n) \cdot \sin(\varphi_n - v_n)$.

Müller & Wehner (1988, PNAS 85:5287-5290) showed that *Cataglyphis fortis* does **not** perform exact trigonometric vector summation; instead it uses a computationally cheap **approximation** — replacing the sine and cosine in the update with piecewise-linear (cosine) and low-order-polynomial (sine) approximants plus a scaling factor. Because these approximants are exact only when the animal moves parallel or antiparallel to its home vector ($\varphi - v = 0$ or $\pm\pi$), the running angle is systematically mis-incremented for oblique headings, producing the small, direction-dependent navigational errors observed in classic two-leg (L-shaped) channel experiments (legs of order 5-10 m; walking speed ≈ 0.33 m/s). This is a landmark result: the model is validated precisely *because* it reproduces the animal's systematic errors.

The contrasting **geocentric Cartesian (GC) vector summation** of Mittelstaedt & Mittelstaedt (1973, 1980) accumulates components $x \leftarrow x + \delta \cdot \cos \varphi$, $y \leftarrow y + \delta \cdot \sin \varphi$ (sums of sines and cosines of *absolute* headings); the home vector is the negative resultant $G = -(x, y)$. Because there is no division by r and no difference-angle, the GC scheme has no systematic angular error and no singularity at the nest — it is the "true" vector summation that Müller & Wehner contrast the ant's algorithm against.

For naïve migratory birds, **Mouritsen & Mouritsen (2000, J Theor Biol 207:283-291)** modeled the **clock-and-compass / vector navigation** strategy as a directed random walk: each night's heading is drawn independently from a **wrapped Cauchy** distribution $f(\theta; \mu, \rho) = (1/2\pi) (1 - \rho^2) / (1 + \rho^2 - 2\rho \cos(\theta - \mu))$ centered on the inherited migratory direction μ , with concentration $\rho = r_{\text{step}}$. Summing n unit step-vectors and using mathematical-expectation theory, they proved the expected geographic spread of recoveries grows as a **parabola** in migratory distance. Fitted to ringing recoveries, $r_{\text{step}} \approx 0.665$ for Scandinavian Pied Flycatchers (*Ficedula hypoleuca*, $\approx \pm 52^\circ$ angular SD) and $r_{\text{step}} \approx 0.7$ best-fit both flycatchers and European Robins (*Erithacus rubecula*),

with a step length of 125 km; the model predicted 66% of recoveries inside the parabola and 65% were observed there. Because headings are independent across nights, errors accumulate without correction — the diagnostic signature of pure vector navigation (no map), consistent with displacement experiments (Perdeck 1958) where naïve birds fail to compensate. [ScienceDirect](#)

[ResearchGate](#)

(c) Validation / species. Desert ants are the canonical model (Wehner and colleagues, decades of fieldwork); the path integrator runs continuously and is "reset" at the nest — displaced ants run off their home vector even away from the nest (Knaden & Wehner 2006). Honeybees show analogous path integration. Bird clock-and-compass is validated against large ringing-recovery datasets and laboratory orientation (Emlen-funnel) tests. [EurekaMag](#)

(d) Software / references. Mittelstaedt & Mittelstaedt (1973, 1980); Müller & Wehner (1988, PNAS); Hartmann & Wehner (1995, Biol Cybern) — neural architecture; Wehner reviews; Mouritsen & Mouritsen (2000, J Theor Biol); Mouritsen (1998, Anim Behav 56:899–907; 2018, Nature 558:50–59 review); Vickerstaff & Cheung (2012, J Theor Biol) — coordinate-system comparison. These are mostly bespoke models rather than packaged software.

(e) Limitations. Path integration accumulates error unboundedly without external correction, so it must be combined with landmark/visual guidance (hence cue integration, Model 6). The Müller-Wehner approximation is deliberately imperfect. The bird clock-and-compass model is debated: Thorup, Rabøl & Madsen (2000) argued recoveries are slightly *more* concentrated than the simplest model predicts (best fit $r \approx 0.72$), implying some on-route error correction. The models describe heading generation, not the full movement-with-error process that SSMs capture. [Macroecointern](#)

Model 6 — Cognitive and Learning Models (RL, Successor Representation, Bayesian Integration, GP Routes)

(a) Core idea. These models answer "where does the animal *decide* to go?" rather than "how does it move?" They range from value-based reinforcement learning and the hippocampal cognitive map, through Bayesian optimal cue integration, to data-driven Gaussian-process descriptions of learned habitual routes. Their relationship to Models 1–5 is that they *generate the drift/bias field* that the mechanistic models take as a given input.

(b) Key mathematics. *Reinforcement learning*: the agent learns a value function over states. Temporal-difference learning updates $V(S_t) \leftarrow V(S_t) + \alpha[R_{t+1} + \gamma V(S_{t+1}) - V(S_t)]$. The control versions are SARSA (on-policy): $Q(S_t, A_t) \leftarrow Q(S_t, A_t) + \alpha[R_{t+1} + \gamma Q(S_{t+1}, A_{t+1}) - Q(S_t, A_t)]$, and Q-learning (off-policy), which replaces the next-action term with $\max_a Q(S_{t+1}, a)$. Action selection uses ϵ -greedy or softmax (Boltzmann) policies. A softmax policy over a learned value landscape is a biased random walk: the value gradient ∇V acts as the drift, and the softmax temperature sets the diffusion — directly linking RL to the BCRW/Langevin picture (Sutton & Barto, *Reinforcement Learning: An Introduction*).

Successor representation (SR): instead of caching values, the agent caches expected discounted future state occupancy, $M(s,s') = E[\sum_t \gamma^t 1(S_t=s') \mid S_0=s]$. Then $V = M \cdot R$, so value is a linear readout of a predictive map. Stachenfeld, Botvinick & Gershman (2017, *Nature Neuroscience* 20(11):1643–1653, "The hippocampus as a predictive map") argue place cells encode rows of the SR and that "entorhinal grid cells encode a low-dimensional basis set for the predictive representation, useful for suppressing noise in predictions" — explaining policy- and topology-dependence of place fields that a purely spatial map cannot. This connects RL to the O'Keefe & Nadel cognitive-map tradition (place cells; Tolman's latent learning; grid cells, Hafting et al. 2005).

Bayesian cue integration: when combining independent noisy cues (path integration vs landmarks; sun vs wind compass), the optimal estimate is the inverse-variance-weighted average; the combined estimate has lower variance than any single cue. Wystrach, Mangan & Webb (2015, *Proc R Soc B* 282:20151484, "Optimal cue integration in ants") showed that *Cataglyphis velox* "show progressively stronger weighting of their PI direction as PI length increases," and that "the weighting is quantitatively predicted by modelling the expected directional variance of home vectors of different lengths and assuming optimal cue integration." In dung beetles, weighted vector-sum integration of sun and wind cues matches the same reliability-weighting prediction.

Gaussian-process route models: Mann et al. (2011, *J R Soc Interface* 8:210–219) treated a homing pigeon's repeated flights as draws from a Gaussian process — each flight an imperfect attempt to recapitulate a latent idealized route, with covariance kernel encoding spatial correlation. The GP's predictive power grows with experience (mirroring learning), and reduced-rank GP approximations objectively identify the discrete waypoints/landmarks the bird relies on. The model found habitual routes are best described as sequences of isolated waypoints rather than continuous paths. ([Washington University in S...](#))

(c) Validation / species. RL and SR are validated primarily against rodent hippocampal recordings (place/grid cells) and behavior (Morris water maze analogues), plus a large machine-learning literature. Bayesian cue integration is validated in desert ants, bees, and dung beetles. The GP route model is validated on GPS tracks of homing pigeons (*Columba livia*), the canonical avian navigation model species.

(d) Software / references. Sutton & Barto (2018); Dayan (1993) — SR; Stachenfeld, Botvinick & Gershman (2017, *Nat Neurosci*); Gershman (2018, *J Neurosci*) — SR review; O'Keefe & Nadel (1978); Tolman (1948); Wystrach et al. (2015, *Proc R Soc B*); Hoinville & Wehner (2018, *PNAS*) — optimal multiguideance integration; Mann et al. (2011, *J R Soc Interface*).

(e) Limitations. RL/SR are largely models of *neural representation and decision*, with only an indirect, often qualitative, link to observed movement statistics — they are rarely fit to free-ranging trajectory data the way Models 1–4 are. Bayesian cue-integration models assume the animal has access to (and correctly weights) cue reliabilities, which is hard to verify. GP route models are descriptive/predictive rather than mechanistic — they quantify route fidelity beautifully but do not explain the underlying navigational computation.

Briefly noted (the next tier)

- **Ornstein-Uhlenbeck position process** for home ranges (Blackwell 1997): the mean-reverting SDE $dz = -\beta(z-\mu)dt + \sigma dW$ gives a Gaussian stationary home range; the foundation of the ctmm OU/OUF family.
- **Advection-diffusion / Fokker-Planck PDE models** of population redistribution (Okubo & Levin; Turchin 1998): the Eulerian/population-level limit of the Lagrangian/individual random walk, with diffusion coefficient D estimable from the MSD.
- **Agent-based / individual-based models** (e.g., Couzin et al. collective motion) for emergent group navigation and "many wrongs" effects.
- **Information-theoretic route-fidelity measures** (positional entropy; Mann/Freeman pigeon work) for quantifying learning and predictability of repeated routes.
- **The movement ecology paradigm** (Nathan et al. 2008, PNAS 105:19052-19059): the organizing framework that decomposes any movement path into internal state ("why move?"), motion capacity ("how to move?"), navigation capacity ("where to move?"), and external factors — the conceptual scaffold these six model families instantiate.

Synthesis: The Family Tree of Navigation Models

The six families are not competitors but layers of one hierarchical model. Read along four axes:

1. Discrete time $\leftrightarrow$ continuous time (and the diffusion limit). The discrete CRW (Model 1) is the Euler-Maruyama discretization of a continuous-time velocity-jump process; the CTCRW (Model 3) is exactly its continuous-time limit, with the OU velocity SDE replacing the step/angle pair. Taking the diffusion (small-step, many-step) limit and invoking the CLT converts either into the Fokker-Planck/advection-diffusion PDE that governs population redistribution — the Eulerian view. So "individual step $\rightarrow$ whole-trajectory distribution $\rightarrow$ population density" is one continuous chain, and the Kareiva-Shigesada MSD is the bridge that lets you read the diffusion coefficient off individual paths.

2. Movement-generation $\leftrightarrow$ space-use estimation. Models 1, 2, 5 *generate* trajectories (forward, mechanistic). Models 3 and 4 *estimate* parameters and space-use distributions from data (inverse, statistical). The link is that the stationary distribution of a generative movement model is the utilization distribution that the estimation models target: an OU position process $\rightarrow$ Gaussian home range (AKDE); an SSF/BCRW with habitat drift $\rightarrow$ a Boltzmann/Gibbs utilization distribution $\propto \exp(\beta \cdot x)$. The SSF is precisely the device that welds the CRW movement kernel (Model 1) to environmental drivers and yields that space-use distribution.

3. Mechanistic ("how it moves") $\leftrightarrow$ cognitive ("where it decides to go"). The BCRW/SSF bias term, the path-integration home vector, the RL value gradient, and the Bayesian posterior mean over the goal are *the same object at different levels of description*. An RL softmax policy over a value function generates a biased random walk whose drift is ∇V ; a path integrator supplies a deterministic home-vector drift; Bayesian cue integration computes the drift direction as a

reliability-weighted average of compass estimates. In every case, Model 6 (decision) supplies the bias that turns the Model 1 CRW into a BCRW. The successor representation makes this especially clean: it is a predictive map (cognitive) whose linear readout is a value field whose gradient is a drift (mechanistic).

4. Latent structure: state-switching and hierarchy. HMMs (Model 3) generalize the single CRW by letting a latent behavioral state index *which* CRW is active, with covariate-driven transitions — subsuming Morales et al.'s mixture-of-random-walks and naturally accommodating the multi-mode behavior (e.g., path integration vs landmark guidance) that cue-integration models (Model 6) describe. Add a measurement-error layer and you have the full state-space model that contains essentially all the others as special cases: choose the transition density to be a biased correlated random walk, let its drift be a value/posterior field, let its parameters switch with a latent state and depend on habitat, and take the continuous-time/diffusion limit when you want population redistribution.

The single sentence a Bayesian/RL-trained reader should carry away: **the entire literature is inference over (and simulation from) a state-space model whose transition kernel is a habitat-modulated, state-switching biased correlated random walk, whose drift field is a policy/value gradient or Bayesian posterior over a navigational goal, and whose continuum limit is an advection-diffusion PDE.**

Recommendations

If your goal is to model observed trajectories (telemetry/GPS):

1. *Start* with an HMM in moveHMM/momentuHMM if sampling is regular and you care about behavioral states; switch to crawl's CTCRW (or state-switching CTCRW) if sampling is irregular or measurement error is large. Benchmark: if residual autocorrelation in pseudo-residuals is high or AIC strongly prefers more states, escalate model complexity.
2. *For home range / space use*, use ctmm's OUF + AKDE, not conventional KDE — Noonan et al. (2019) show IID estimators underestimate area by a factor of ~2 to 13 when locations are autocorrelated. Use a dynamic Brownian bridge instead only when you want the occurrence (path) distribution, not the range distribution.
3. *To link movement to environment*, fit an iSSA in amt; include $\log(\text{step length})$, step length, and $\cos(\text{turn angle})$ so you recover the movement kernel, and add $\text{habitat} \times \text{movement}$ interactions. This is your BCRW-with-habitat.

If your goal is to test a navigational *mechanism*: 4. For homing/goal-directed data, fit a BCRW and estimate the bias weight β (Bailey et al. 2018 parameterization); $\beta \approx 0$ means no detectable map-and-compass, large β means strong goal orientation. Guard against the persistence-mimics-bias confound (Benhamou 2006) by using individuals with known, varied goal directions. 5. For insects/path integration, fit GP or GC vector-accumulation models and *test for systematic errors* (the Müller-Wehner signature) in two-leg/displacement experiments — error structure

discriminates the approximate polar from the exact Cartesian integrator. 6. For naïve migrants, fit the wrapped-Cauchy clock-and-compass model and test the parabolic-spread prediction against recovery/tracking data; deviation toward *over*-concentration (as Thorup et al. found) is evidence of on-route error correction and argues for adding a map/cue-integration layer.

If you are bringing an RL/Bayesian toolkit: 7. Treat the mechanistic models as the generative/likelihood layer and your RL/Bayesian models as the prior over the drift field. Concretely: parameterize the BCRW/SSF drift as ∇V where V is a learned value function or a Bayesian posterior over goal location, and fit end-to-end. The successor representation is the natural predictive-map prior. 8. *Avoid* the Lévy-walk trap: only claim a Lévy process after maximum-likelihood fitting against truncated-power-law, exponential, gamma, *and* composite-CRW alternatives with AIC; a power-law fit alone is not evidence of a Lévy mechanism.

Thresholds that should change your approach: irregular sampling or large location error → continuous-time SSM; multimodal step/angle histograms → add HMM states; strong habitat gradients → iSSA; evidence of route stereotypy across repeats → GP route model + positional entropy; systematic homing errors → approximate path integrator; unbounded error accumulation → add Bayesian cue integration.

Caveats

- **Selection of "top 6" reflects adoption, not correctness.** These were chosen by citation weight, canonical-package availability (moveHMM, momentuHMM, crawl, ctm, amt), and foundational status (Kareiva-Shigesada, Codling et al., Nathan et al.). Other defensible groupings exist; the Lévy family is included for its outsized influence despite being the most contested.
- **Scale dependence is pervasive.** CRW and HMM parameters depend on the observation interval; cross-study comparisons of step/angle parameters are only valid at matched sampling resolutions.
- **The Lévy literature carries a documented history of statistical error.** Treat pre-2008 Lévy claims with caution; many were overturned by the maximum-likelihood reanalyses of Edwards and colleagues, and composite CRWs reproduce apparent power laws.
- **The bird clock-and-compass model is actively debated.** Mouritsen & Mouritsen (2000) and Thorup et al. (2000) disagree on whether recovery distributions imply pure vector navigation or partial error correction; some species clearly possess a true map sense that the simple model omits.
- **The exact coefficients of the original Müller-Wehner (1988) approximation could not be verified verbatim** from the primary PNAS text within this review; the *structural form* (piecewise-linear cosine + polynomial sine + scaling, producing direction-dependent errors) is firmly established across secondary reviews (Merkle et al. 2006; Vickerstaff & Cheung 2012), but anyone re-implementing it should confirm the numerical constants against the original paper or Hartmann & Wehner (1995).

- **Cognitive models (Model 6) are the least directly fit to movement data.** RL/SR are validated mainly on neural recordings and controlled tasks; their connection to free-ranging trajectory statistics is conceptual and an active research frontier rather than established practice.
- **Where sources conflict** (e.g., optimal Lévy exponent in >1 dimension; pure vs corrected vector navigation), this report flags the disagreement rather than adjudicating it.